

RESEARCH ARTICLE

Optimizing ResNet-50 for Multiclass Classification: A Multi-Stage Learning Approach

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ABSTRACT In this study, we present a multistage learning pipeline that utilizes the ResNet-50 architecture as a static feature extractor for multiclass image classification problems. This methodology integrates transfer learning, data augmentation, and adaptive learning techniques to enhance generalization across unbalanced and diverse datasets. We evaluated our approach using HAM10000, CIFAR-10, and CIFAR-100 to indicate its impact in both the medical and natural image domains. In contrast to providing a new network architecture, our contribution highlights a realistic and reproducible training schedule that compares effectively with present soft models such as EfficientNet-V2 and MobileViT-v2. Experimental results validate that our pipeline provides strong classification performance with minimal CPU resources, underscoring its applicability to practical image classification applications.

INDEX TERMS CNNs, data augmentation, deep learning, dataset imbalance, multiclass classification, ResNet-50, transfer learning.

I. INTRODUCTION

Multiclass classification represents a significant challenges in machine learning, including the assignment of input data to one of many predetermined categories. The issue is prevalent across various domains, including medical imaging, where the precise categorization of skin lesion types from dermoscopic images facilitates early diagnosis, and natural language processing in applications such as topic classification and sentiment analysis. As datasets expand in size and complexity, efficient solutions are necessary to manage high-dimensional feature spaces and multiple-class distributions. Deep learning techniques, particularly Convolutional Neural Networks (CNNs), have demonstrated considerable efficacy in image classification due to their capacity to autonomously generate hierarchical features from raw data. Among these models, ResNet-50 has gained prominence due to its substantial architecture and the integration of residual connections, which mitigate the vanishing gradient problem and enhance the efficient training of deeper networks. ResNet-50 has demonstrated superior performance in several

applications, including medical image analysis and object identification, with simpler architectures in terms of accuracy and efficiency. Despite its advantages, ResNet-50, similar to several deep learning models, may encounter difficulties in multiclass classification tasks when faced with unbalanced datasets or subtle class distinctions. We propose a divided classification methodology to address these difficulties. Initially, binary classification was used to partition the intricate multiclass issue into simpler subproblems, thereby explaining the decision limits. In the second phase, ResNet-50 was utilized as the primary architecture, focusing on its profound feature extraction skills to enhance classification precision. This technique resembles transfer learning but is distinct in that it employs a multi-stage pipeline instead of direct class mapping. A previous study investigated several methodologies for enhancing multiclass classification, including conventional machine learning models and contemporary CNN-based architectures. However, constraints remain, particularly in the context of extensive datasets and complex interclass connections. This study aims to enhance the field by evaluating the efficacy of a binary-to-multiclass pipeline utilizing ResNet-50, specifically addressing issues of class imbalance and complex category differences. We performed

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our tests on three benchmark datasets: HAM10000 (a dermoscopic image dataset extensively utilized for skin lesion detection), CIFAR-10, and CIFAR-100. CIFAR-10 has 10 object classes, whereas CIFAR-100 includes 100 classes categorized into 20 superclasses (including “aquatic mammals,” “flowers,” and “vehicles”), with each superclass comprising 5 fine-grained subclasses. The primary objective of this study was to enhance the accuracy of multiclass classification by integrating binary classification techniques with the architectural benefits of ResNet-50. Furthermore, we explore how strategies such as addressing class imbalance and adapting datasets might enhance performance across many domains.

This study analyses the following research questions:

- 1 -How can the combination of binary classification and ResNet-50 improve the multiclass classification accuracy in datasets with complex class relationships and imbalances?
- 2 -How can the integration of dataset-specific techniques and class-imbalance management enhance the performance of ResNet-50 in diverse classification tasks

This paper is arranged as follows: Section II defines the previous studies about multiclass classification method and datasets. Section III defines the study’s methodology, specifying the strategy and experimental configuration utilized. The experimental results, presents in Section IV followed by Section V, which contains the discussion. Section VI concludes this study and describes the prospects for future work.

II. LITERATURE REVIEW

Reference [1] this research indicates that the complexity of multidimensional functional data, characterized by infinite-dimensional characteristics, renders traditional categorization methods inadequate. The authors provide a distinctive multi-class functional deep neural network (mfDNN) classifier to tackle the complexities of multidimensional functional data. This classifier employs a streamlined deep neural network architecture with a rectified linear unit (ReLU) activation function, aimed at minimizing cross-entropy loss in a multiclass classification context. In this research, proposed, the base technique on a solid theoretical foundation that offers convergence rates for misclassification risk functions for both completely and discretely seen data. Moreover, in their research, they implement the mfDNN classifier utilizing advance computer resources, demonstrating its practical applicability. The technique exhibits effectiveness on both simulated datasets and diverse benchmark datasets across several application fields. This paper addresses the limitations of existing classification methods for multidimensional functional data and concurrently synthesizes the techniques for one-dimensional and multidimensional classification challenges. The proposed solution signifies a significant progression in the domain. It employs multidimensional functional data rather than independent and

identically distributed (i.i.d.) data, which is the basis for most multiclass classification algorithms. This document presents foundational research, offering a theoretical framework for multiclass classification in multidimensional functional data and creating a supplementary benchmark in the field [2] This study determined the effectiveness of a multiclass deep learning model for classifying skin lesions using both image data and textual descriptions, including patient demographics like age, sex, and lesion location. The experiment apply the HAM10000 dataset, comprising 10,000 images of skin lesions. They assessed the multimodal approach against conventional deep learning systems using solely image data. The results showed that the multimodal model using the Align before Fuse (ALBEF) architecture exceeded the other models with an accuracy of 94.11% and an AUCROC of 0.9426. Including more data kinds—such clinical information—with image data helps to improve the classification of skin lesions, therefore emphasizing the advantages of doing so. Additionally compared the performance of the multimodal model with those of well-known deep learning systems including InceptionV3, ResNet50, and DenseNet121, which showed higher accuracy and AUCROC values. By combining a Vision Transformer for picture encoding and BERT for text encoding, the ALBEF model shows great promise for future applications in primary care, therefore improving the dependability and accuracy of skin cancer screening by medical professionals. Furthermore, including textual data as a supplementary feature in classification tasks enhances the understanding of skin lesions, therefore making the model suitable for practical healthcare applications. This study advances the use of multimodal deep learning in medical image analysis and creates a basis for integrating a wider array of information in future research to improve model effectiveness. Although the good outcomes, the article recognizes numerous limitations, including the restricted metadata utilized in the experiments, and proposes that the integration of more textual content might enhance the categorization findings [3] Imbalanced datasets sometimes lead classifiers to favor the dominant class, hence restricting the precise identification of minority classes. Although previous studies have mainly concentrated on rectifying imbalances in binary classification, the concern raises considerably in multiclass contexts. In these cases, an imbalance may result in issues such as inadequate recognition of minority class occurrences, data bias, and model overfitting. Traditional methods have attempted to resolve this by either transforming the problem into several binary categories or employing generalized solutions that persistently encounter overlapping difficulties and lower recognition rates for minority classes This research presents a hybrid technique that combines data-level balancing with algorithm-level optimization. The data-level approach enhances the number of minority class samples, while the algorithm-level enhancement creates a cost-sensitive dataset that mitigates model bias towards the majority class. The proposed hybrid method aims to enhance prediction accuracy by focusing on underrepresented classes.

Results show that imbalanced multiclass datasets compared to clear improvement in classification performance existing techniques. Also, even when the number of minority examples is very limited, the performs well in both majority and minority-dominated class scenarios. It has proven particularly useful in handling data imbalance, improving minority class recognition, and reducing the risk of overfitting [11]. This study highlights the efficacy of AI-driven testing techniques in detecting unusual skin lesions. It analyses the use of cosine cyclic learning rate to enhance the accuracy of skin cancer categorization. Three major convolutional neural networks (CNNs) architecture — VGG19, ResNet101, and InceptionV3 — serve as feature extractors. Each network is assessed using three learning rate strategies: constant, cosine annealing, and cosine annealing with warm restart. Experiments using the HAM10000 dataset, including dermatoscopic images of seven categories of skin disease. The findings indicate that the cosine cyclic learning rate with warm restart significantly enhances model performance. Among the evaluated networks, ResNet101 and InceptionV3 had the greatest accuracy rate of 92%, surpassing the baseline models. This underscores the efficacy of the suggested strategy in identifying skin lesions with enhanced precision and efficiency. Research indicates that deep learning models may achieve enhanced diagnostic accuracy by modifying learning rate schedules, hence underscoring their significance in clinical environments for the early detection of skin cancer. This work demonstrates the efficacy of cosine cyclic learning rates in various CNN architectures, hence enhancing our understanding of deep learning applications in medical imaging. These results underscore the need of using advanced optimization approaches in medical systems that use artificial intelligence. Future research will explore advanced CNN architectures and learning methodologies aimed at enhancing diagnostic precision and facilitating early detection, both critical for increasing cancer patients' survival rates [5]. In healthcare, the early discovery of cutaneous cancers is crucial due to their increasing prevalence, and effective treatment relies on prompt diagnosis. Traditional methods for diagnosing skin cancer, such as dermatological examinations and histological analyses, have limitations due to their subjective nature, which may result in wrong diagnosis [6]. The rising influence of machine learning in medical diagnostics has increased interest in using these technologies to enhance the accuracy and dependability of skin cancer detection. Specifically, deep learning methodologies, including Convolutional Neural Networks (CNNs), have shown encouraging outcomes in the detection of skin lesions from microscopic images. Numerous studies have shown that CNNs may, in some instances, surpass dermatologists in the early identification of conditions, offering a reliable and scalable diagnostic option. Notwithstanding their promise, these models often exhibit susceptibility to overfitting, a concern that becomes particularly evident when confronted with limited or unbalanced datasets [7]. Multiple solutions have

been developed to address the difficulties of not adequate data and class imbalance. Data augmentation methods, including image rotation, zooming, and flipping, are often used to expand the training dataset and reduce overfitting consequences [8]. A particularly persistent problem is the disparity between benign and malignant skin lesion categories. This issue has been mitigated using resampling techniques such as the Synthetic Minority Oversampling Technique (SMOTE), which artificially enhances the quantity of minority class samples to achieve dataset equilibrium [9]. These methods have shown that they can improve the performance of models, especially when it comes to finding rare classes, which is important for making a quick diagnosis. Researchers have looked into using both traditional machine learning algorithms and deep learning for skin cancer classification tasks. Support Vector Machines (SVM), Random Forests (RF), and K-Nearest Neighbours (KNN) are examples of classifiers that have had different levels of success [8], [9]. Random Forests are often the best of these because their ensemble mechanism combines predictions from several decision trees, which makes the model more accurate and stable. One interesting study found that Random Forest was 95.6% accurate on the HAM10000 dataset, which was better than models based on single decision trees and KNN. The Ridge Classifier and SVM are two other methods that have also done well on large datasets, with reported accuracies close to 90%. [10] Researchers looked into how to make these models better by combining clinical data with dermoscopic images. According to this research [11], skin cancer classifiers are more accurate when clinical markers such age, sex, and lesion site are combined with image data. This multimodal method, which integrates visual and non-visual information, enables the model to make more precise predictions especially in murky or edging circumstances. Despite these developments, the sector is still evolving and academics are still trying to create more accurate models. Future researchers will probably investigate more complex deep learning systems, such as transformers, as well as methods to improve data preprocessing. Particularly in locations with limited resources, transfer learning—which enhances models created on vast databases for finding skin cancer—holds a lot of promise to increase diagnostic accuracy [12]. Although significant studies in the utilization of machine learning methodologies for skin cancer diagnosis, challenges such as dataset imbalance, model overfitting, and the incorporation of clinical data continue. The integration of conventional machine learning models, such as Random Forest, with deep learning approaches, in conjunction with strategies like data augmentation and resampling, consistently shows the potential to enhance skin cancer diagnosis. Subsequent study will likely concentrate on developing these methods in order to enhance model precision [6]. Recent research has shown that deep learning models, especially convolutional neural networks (CNNs), has significant potential in the categorization of skin lesions. Models such VGG-16, ResNet, and InceptionV3 have been used to

identify skin tumors in dermoscopic images, using big datasets like HAM10000, which has thousands of annotated dermatological images. Nonetheless, difficulties including uneven class distributions, which inadequately represent some skin lesion types, have led to reduced model accuracy, especially for minority classes. Data augmentation and balancing strategies, including as up-sampling, under-sampling, and synthetic minority oversampling techniques (SMOTE), have been utilized to solve these imbalances, often improving the classification accuracy of models. This study's main contribution is the use of a transfer learning approach, especially utilizing ResNet-50, to classify skin cancer lesions. Transfer learning is beneficial when data is scarce, since it enables models to utilize pre-trained weights from extensive datasets, like ImageNet, and adjust them for smaller, more specialized datasets, such as HAM10000. This work included upsampling and data augmentation techniques, resulting in a balanced dataset of 45,844 images and assuring requisite data variety for improved model performance. The suggested methodology utilizes many data augmentation techniques, including rotation, zoom, width shift, height shift, flipping, and fill mode, to improve model generalization and reduce overfitting. These tactics are crucial for enhancing dataset variability and avoiding the model from developing biased patterns that do not generalize adequately to new data. Utilizing the ideal hyperparameters (Adam optimizer, a learning rate of 0.001, a batch size of 64, and 22 epochs), the ResNet-50 model attained a classification accuracy of 98%. This outcome signifies a substantial enhancement compared to prior investigations, which frequently recorded diminished accuracy rates, especially in the context of unbalanced data. Despite the model's demonstration, many limitations continue. The reduction in image size from the typical 224×224 pixels (utilized by ResNet-50) to 150×150 pixels was crucial for improving computing performance, given the large dataset. However, this image scaling may have somewhat reduced the quality of feature extraction and model effectiveness. Subsequent study may concentrate on investigating larger image dimensions and enhancing various elements of the preprocessing processes, including image noise mitigation, to augment classification accuracy. Additionally, techniques like SMOTE, which generate synthetic samples for under-represented classes, can be employed to increase the class imbalance and improve model efficacy for prevalent skin lesions [7]. The present research aims to address this issue by developing an efficient classification model for skin cancer lesions with the optimized EfficientNetV2-B2 architecture. The model was trained and evaluated using the HAM10000 dataset, which comprises 10,015 dermoscopic images of diverse skin lesions. The proposed model was trained using a carefully selected set of hyperparameters: the Adamax optimizer, a learning rate of 0.001, batch size of 64, and image resolution scaled to 260×260 pixels. During preprocessing, data augmentation techniques such as horizontal and vertical flipping were applied to increase variability and enhance training effectiveness. Resizing and normalization

were also used to maintain consistency across input images.

The EfficientNetV2-B2 model achieved strong performance, with an F1-score of 80% and classification accuracies of 89%, 81%, and 80% across different metrics. These results demonstrate the model's capacity to accurately detect skin cancer lesions even in imbalanced datasets. However, the study identifies potential limitations. One notable constraint is the absence of advanced techniques for handling class imbalance—such as upsampling or downsampling—which could further improve the model's generalization across various skin cancer types. The study points to the possibility that the feature variance resulting from the adopted augmentation techniques may not adequately reflect the intricacy of skin cancer lesions, therefore implying that more augmentation methods or feature engineering technologies might improve the model's efficacy. Targeting skin lesion categorization, this study presents a new approach employing fine-tuning methods on the Xception pretrained model. Using the HAM10000 database—an exhaustive collection of dermatological pictures with lesions divided into seven categories—this study was carried out. This study applied various data augmentation techniques—such as zooming, rotation, and flipping—to address the common issue of class imbalance in medical imaging datasets. These methods helped increase dataset diversity and contributed to improved model stability and generalization. By expanding the number of samples, particularly for underrepresented classes, the model becomes better suited for real-world medical classification tasks. Specifically VGG16, ResNet50, DenseNet169, DenseNet201, and Xception are compared in a thorough analysis in the current article on several deep learning models. The models were evaluated using standard performance metrics, including accuracy, precision, recall, and F1-score. Among the tested models, the fine-tuned Xception network delivered the most consistent results across all metrics. It achieved an accuracy of 96%, indicating strong performance in skin lesion classification compared to other architectures. These findings highlight the effectiveness of advanced convolutional neural networks (CNNs), such as Xception, in supporting early and reliable skin cancer diagnosis. Early identification and timely action in dermatology are promoted by accurate estimates of skin lesion categories made possible by leveraging Xception's capabilities. According to the study results, when combined with sophisticated augmentation methods, fine-tuning of deep learning models can greatly improve the performance of artificial intelligence programs in skin cancer diagnosis [9]. This paper introduces a deep learning classification model especially developed for the accurate and rapid detection of skin cancer. The intricacy of image analysis for skin cancer diagnosis is exacerbated by data imbalance in healthcare datasets, where class difference may result in the overfitting of more dominant classes, hence impeding model efficacy. The research employed the HAM10000 (Skin Cancer MNIST) dataset, including 10,015 pictures classified into seven unique categories of skin lesions

The proposed approach applies image preprocessing and enhancement techniques to reduce noise and improve lesion image clarity. Pre-trained networks such as InceptionV3, InceptionResNetV2, MobileNetV2, and EfficientNetB0 are used for classification, with optimizations applied to mitigate overfitting and boost performance. This method achieved an overall accuracy of 81.3%. The study further suggests that model generalizability could be enhanced through techniques such as greyscale conversion and expanded data augmentation. These methods may contribute to reducing overfitting and improving the robustness of deep learning models in skin cancer detection. The findings underscore the growing importance of deep learning in dermatological diagnostics and advocate for future research to incorporate larger datasets and more advanced training methodologies [10]. Another section of the research focuses on evaluating EfficientNet architectures for the classification of dermoscopic images, with a particular emphasis on diagnostic accuracy and computational efficiency. Given the prevalence of skin cancer, developing accurate and timely diagnostic tools is vital. The HAM10000 dataset, comprising 10,015 images across seven distinct skin lesion categories, was utilized along with a range of data augmentation strategies. Among these, flip augmentation demonstrated the highest initial classification accuracy of 80.78%. All five evaluated EfficientNet versions (B0 to B4) consistently achieved accuracy levels above 79.04%. Notably, EfficientNet B3 delivered the best results, reaching 87.75% accuracy following fine-tuning of the convolutional layers—outperforming previous methods applied to the same dataset. These results confirm the effectiveness of EfficientNet B3 in skin lesion analysis, with much of the performance improvement attributed to the synergistic effect of fine-tuning and augmentation techniques [12]. In addition, the study proposes a distributionally robust optimization (DRO) method for multiclass logistic regression (MLR) aimed at enhancing model resilience to data outliers. This approach introduces a confusion set defined via the Wasserstein distance, capturing probability distributions similar to the empirical distribution of the training data. The original min-max problem is reformulated into a regularized learning task, where the regularization component is defined by the norm of the model's coefficient matrix. Both theoretical analysis and experimental validation support the connection between regularization and improved model robustness. When applied to Vision Transformer (ViT)-based image classifiers, the proposed method demonstrated enhanced prediction accuracy in the presence of both random and adversarial perturbations, as evidenced by experiments conducted on the MNIST and CIFAR-10 datasets. The findings indicate substantial enhancements in performance, with test error reductions of 83.5% and loss reductions of 91.3% relative to baseline methodologies. This paper offers theoretical ideas and practical facts that substantiate the effectiveness of the DRO-MLR method in creating robust multiclass classifiers. This study investigates the categorization of external ocular disorders,

including blepharitis, cellulitis, and conjunctivitis, utilizing a specialized deep Convolutional Neural Network (CNN) model for multiclass classification. The model is built on the VGG-16 architecture and evaluated with dropout rates of 0.25 and 0.50 to examine their impact on classification performance. After preprocessing, the dataset was split into training (80%, 968 samples) and testing (20%, 242 samples) sets. The deep CNN achieved a classification accuracy of 98.48%. For the multiclass task, 10-fold cross-validation yielded R^2 values ranging from 0.425 to 0.775. Specifically, VGG-16 recorded 71.54% accuracy when a dropout rate of 0.50 was applied. Applying VGG-16 to external eye diseases in a multiclass setting with variable dropout rates is considered a novel approach, as such a configuration has not been previously studied [14]. The study also introduces a quantum machine learning method that uses Quantum Convolutional Neural Networks (QCNNs) for multiclass classification, a task relevant to areas like computer vision. A hybrid quantum-classical framework is implemented through TensorFlow Quantum, integrating quantum computing with classical optimization strategies. The quantum outputs are processed using a softmax activation function to minimize cross-entropy loss and optimize circuit parameters. The model architecture incorporates a quantum perceptron and a redesigned quantum circuit to enhance classification accuracy. It is tested on a four-class version of the MNIST dataset, which contains handwritten digits. Eight qubits are used for data encoding, along with four ancilla qubits to support circuit complexity. Unlike previous studies that primarily addressed three-class classification tasks, this setup expands the scope by applying quantum convolutional neural networks (QCNNs) to more complex scenarios. The QCNN model achieved accuracy levels comparable to classical convolutional neural networks, despite using a similar number of trainable parameters. These findings suggest that quantum neural networks could offer a viable alternative for multiclass classification, especially as quantum hardware continues to improve. Nonetheless, the study notes that current implementations require multiqubit gates, and improvements in quantum processors may lead to more efficient training and better overall performance. These processors may enable the development of more scalable and resource-efficient quantum algorithms, improving their relevance to practical classification problems, particularly in the Noisy Intermediate-Scale Quantum (NISQ) era, marked by continuing developments in quantum technology [16]. Multi-label classification is a fundamental challenge in numerous application domains, where a single image or instance can belong to multiple categories. This study proposes a multi-branch neural network (MBNN) architecture that enables input encoding through semi-parallel subnetworks. The proposed architecture is optimized for multitask learning (MTL) efficiency, particularly in multi-label classification scenarios. The framework employs convolutional neural networks (CNNs) as subnetworks, with each subnetwork dedicated to extracting features for distinct label

predictions. Two configurations—multi-output and multi-feature networks—were implemented using transfer learning from pre-trained CNN models. Experimental results indicate that the multi-feature architecture outperformed alternatives, attaining an F1-score of 83.31% on the Amazon Rainforest dataset. Consistent performance was observed across benchmark datasets, with scores of 88.81% (Pascal VOC 2007), 87.71% (Nuswide), and 88.64% (Pascal VOC 2012). The design mitigates reliance on extensive annotation while maintaining robustness in diverse multi-label contexts. Findings highlight the advantages of heterogeneous CNN output types and feature representations. Hyperparameter optimization further improved classification metrics. Statistical significance was validated via Friedman tests with Nemenyi post-hoc analysis ($\alpha = 0.05$), supported by critical difference (CD) diagrams for pairwise comparisons. Parameter sharing strategies were also investigated, revealing that soft parameter sharing enhanced performance in complex multi-label structures, particularly in multi-feature networks, relative to hard sharing approaches [17]. This study further investigates multiclass classification for multidimensional functional data, which comprises function-valued observations spanning multiple domains. Due to the inherent infinite-dimensional structure of such data, conventional classification techniques often exhibit suboptimal performance. To address this limitation, the authors propose a novel multiclass functional deep neural network (mfDNN) model, which enhances deep learning-based analysis of functional data. A key innovation of the proposed approach involves employing a sparse deep neural network architecture with rectified linear unit (ReLU) activation functions. This design reduces cross-entropy loss in multiclass scenarios, enabling more effective modeling of high-dimensional functional data complexities. The sparse architecture not only improves computational efficiency but also enhances generalization across diverse data types. The method is supported by theoretical guarantees, including convergence analysis of misclassification risk functions. The study's findings remain valid for both fully and partially observed functional data. Theoretical analysis confirms the method's robustness in handling complex multidimensional functional datasets, even when standard independent and identically distributed (i.i.d.) assumptions are violated. To evaluate the mfDNN classifier's efficacy, the authors performed extensive experiments using synthetic and real-world benchmark datasets from multiple application domains. These evaluations covered both one-dimensional and multidimensional functional classification tasks, consistently demonstrating the model's strong performance across varied multiclass classification scenarios [19]. Deep Convolutional Neural Networks (CNNs) have established themselves as fundamental tools for image classification tasks, owing to their exceptional hierarchical feature extraction capabilities. However, their performance typically degrades significantly when trained on limited annotated datasets, often resulting in model overfitting. This investigation systematically evaluates multiple CNN architectures and implements several

methodological improvements to mitigate this limitation. The proposed approach incorporates a domain adaptation framework to better capture dataset-specific features, while optimized activation functions, refined weight initialization strategies, enhanced regularization techniques, and carefully tuned simulation parameters collectively work to minimize overfitting risks. Experimental validation was conducted across three heterogeneous remote sensing datasets (RS19, UC Merced, and EuroSat), with particular emphasis on improving generalization under data scarcity conditions. The framework capitalizes on transfer learning from pre-trained models, thereby dramatically reducing dependency on large labeled datasets. This approach demonstrates particular efficacy in optical remote sensing image classification, where annotated data availability is frequently constrained. Furthermore, comprehensive data augmentation protocols were implemented to increase training sample diversity and enhance model resilience. The synergistic combination of transfer learning and strategic data augmentation yields measurable performance gains in remote sensing classification tasks, as demonstrated by our experimental results [20]. Furthermore, this research tackles two critical challenges in deep neural network (DNN) applications: class imbalance and heterogeneous classification difficulty. These issues frequently result in disproportionate recall rates among different categories, thereby compromising overall model efficacy. To address these limitations, we propose an innovative multiclass minimax framework that replaces traditional output layers with a hybrid decision system combining K-means clustering with minimax Bayes classification. This architecture demonstrates three key advantages: (1) it achieves more balanced recall rates across all classes, (2) effectively processes both imbalanced and non-i.i.d. data distributions, and (3) maintains computational tractability through efficient subgradient optimization methods. The proposed system's design ensures robust performance while preserving the computational efficiency essential for practical deployment [21]. The classification of rarely similar tumors in histopathological images is another area of the research, and it is hindered by uniform cellular patterns and color gradients. In order to solve this problem, the authors created a novel deep neural network (DNN) model integrated with semantic segmentation to allow extraction of important visual details needed for accurate tumor recognition. Its accuracy on a big dataset reached 98.7%, making it useful and trustable for clinical diagnostics of breast cancer [23]. Moreover, the work investigates skin lesion classification using the HAM10000 dataset consisting of dermoscopic images depicting various skin lesions. The approach uses several pre-trained CNN models, with VGG16, MobileNet, and Inception V3 being the most notably used for classification into seven skin categories. Out of the models tested, MobileNet performed best with an 80% test accuracy. Inception V3 and VGG16 followed with 76% and 73%, respectively. For greater clinical utility, the networks were implemented on a web-based platform developed in Streamlit which broadened the scope

TABLE 1. Literature review compration.

Study/Meth od	Strategy	Key Advantages	Limitations	Relevance to Two- Step Approach
ResNet-50 for Multiclas s Classifica tion	CNN- based model with ResNet- 50 as the backbone	Improved accuracy on complex datasets; compact feature extraction from images	Requires large datasets for training; computatio nally challenging	Really relevant as it uses ResNet- 50 in a two-step classifica tion framework.
Hybrid CNN and LSTM Model for Sentiment Analysis	Combine CNN for feature extraction and LSTM for sequence processin g	Improves accuracy in classificatio n of text and sentiment data such as input	May find difficulties with sensitive sequences, increased processing resources	This hybrid methodol ogy may be expanded into the Two- Step Approac h, for feature and sequence modeling
DenseNet for Small- Scale Image Classifica tion	Uses DenseNet architectu re for efficient feature extraction with fewer parameter s	Optimal for limited datasets, minimizes number of parameters while retaining performance	Performanc e can decrease when reality bias distorts, consuming high compute	DenseNe t may be utilized as a feature extractio n technique inside the Two- Step Approac h
Transfor mer- Based Model for Time- Series Classifica tion	Uses transform er architectu re with attention mechanis ms to process time- series data	Cutting- edge performance ; efficient resolution for sequential data	Significant computatio nal resources; requires multi-stage training for max performanc e	Extremel y relevant for time- series tasks in the Two- Step Approac h, particular ly in

TABLE 1. (Continued.) Literature review compration.

				classifica tion
Mixing Strategy for Multiclas s Classifica tion	Applies a mixing or ensemble approach to CNN for improved multiclass classificat ion	Optimizes performance across different classes, better accuracy for underrepres ented classes	May involve significant preprocessi ng steps; poor performanc e on highly unbalanced data	The significa nce is in how these models can be integrate d into the Two- Step Approac h for balanced classifica tion
Tumor Classifica tion with Semantic Segmenta tion and DNNs	Uses semantic segmentat ion combined with deep neural networks for tumor detection	Greater accuracy in tumor detection within medical imaging; reduces errors for finer detection	Requires large datasets; longer training time and complexity	Tumor classifica tion can be included in the Two- Step Approac h for specializ ed medical image tasks
Multi- Class Skin Disease Classifica tion Using Pre- trained CNNs	Transfer learning with pre- trained CNNs for skin disease classificat ion	Efficient solution for distant, limited datasets; fast learning from pre- trained weights	Dependent on quality of source data; performanc e plateaus on difficult targets	Transfer learning and fine- tuning is a fundame ntal step in most Two- Step Approac hes

TABLE 1. (Continued.) Literature review compration.

Semi-supervised Learning for Multiclass Classification with Pseudo-Labeling	Uses pseudo-labeling and semi-supervised learning to classify data with limited labels	Improves performance with limited labeled data; reduces the need for manual labeling	Pseudo-labeling can create noise, errors; sensitive to semi-reliable information	May be included in the Two-Step Approach to help with low-label or sparse data
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of evaluating skin diseases. The study underscores the role of transfer learning in low data scenarios and refined the concept of automated diagnosis in poorly resourced healthcare systems. They intend to expand the work focusing on mobile adaptation for remote and underserved populations [24]. The concluding part of the study focuses on addressing the challenge of limited labeled data. Rather than relying on conventional one-vs-all pseudo-labeling frameworks, the authors adopted a multi-class classification approach that combines transfer learning with fine-tuning. This strategy removes the necessity for oversampling techniques. By applying strict confidence thresholds during pseudo-label assignment, the method enhances computational efficiency and minimizes storage needs, while maintaining strong classification performance—achieving 95.59% accuracy on the MNIST dataset and 84.84% on Fashion MNIST. This presents a scalable and practical pathway for semi-supervised learning in image classification domains [24]. Additionally, the study introduces a distributionally robust optimization (DRO) framework for multiclass logistic regression to mitigate the impact of data outliers. Using the Wasserstein distance, the approach defines a probabilistic ambiguity set that captures distributional uncertainty around the empirical training data. Reformulating the optimization problem as a regularized learning objective—where the regularization term is based on the norm of the model parameters—yields improved generalization. Experiments with Vision Transformer (ViT)-based classifiers on MNIST and CIFAR-10 demonstrate substantial improvements, with test error reductions of up to 83.5% and loss decreases reaching 91.3% compared to standard methods. These findings offer both theoretical justification and empirical validation, revealing a significant connection between robustness and regularization in high-dimensional multiclass learning.

III. METHODOLOGY

This section indicates the methods used for multiclass classification on the HAM10000 dataset, followed by its application to other datasets, including CIFAR-10 and CIFAR-100. The technique comprises data preparation, model design, training strategy, and evaluation measures.

A. DATASETS AND PREPROCESSING

1) HAM10000 DATASET

The HAM10000 dataset contains 10,015 dermoscopic images in seven diagnostic categories. The preprocessing steps for this dataset consisted of resizing, we resized the images to pixels to meet the input requirements of ResNet-50. Normalization of pixel values were scaled accordingly. Data Augmentation: To improve generalization and increase data diversity, We used rotation, vertical flipping, and zooming. Class Balancing: We applied targeted data augmentation to underrepresented classes, ensuring a more balanced class distribution without introducing synthetic samples through oversampling algorithms.

2) CIFAR-10 DATASET

The CIFAR-10 dataset has 60,000 images, categorized into 10 classes, with 6,000 images allocated to each class. Every class contains 500 training photos and 100 testing images. The preparation stages were scaling the images to a consistent dimension, normalizing using the dataset’s mean and standard deviation, and implementing data augmentation techniques like random cropping, flipping, and color buzzing to generate diverse situations and enhance model resilience.

3) CIFAR-100 DATASET

The CIFAR-100 dataset has 60,000 images distributed across 100 classes, with 500 training images and 100 testing images for each class. The preprocessing procedures involved: We adjusted the size of the images. Normalization: Standardization was implemented utilizing the dataset’s mean and standard deviation. Data Augmentation: Random cropping, flipping, and color disruption were used to duplicate varied settings.

B. MODEL ARCHITECTURE

Using ResNet-50 as the main backbone for multiclass categorization across all datasets, the suggested classification pipeline adopts a two-stage strategy. To distinguish between benign and malignant classes, a first binary classifier has been applied to the HAM10000 dataset. The results are then sent to a fine-tuned ResNet-50 model with a custom fully connected (FC) layer created for the precise class count of the data collection. For CIFAR-10 and CIFAR-100, the same ResNet-50 architecture is used, with the final classification layer changed appropriately for the 10 and 100 output classes. To guarantee model flexibility while preserving general-purpose feature representations, the ResNet-50 backbone is first given pre-trained ImageNet weights. While the ultimate residual block is unfrozen and improved during the training process, the first layers remain frozen. By means of this focused fine-tuning, the model can adapt to patterns particular to the dataset and so reduce the chance of overfitting connected with entire retraining. A custom classifier block of dense layers followed by softmax activation replaces the conventional fully linked output layer. This setup

lets the building keep a lightweight weight while improving classification accuracy across several fields. To assess the effect of freezing and fine-tuning settings on classification performance, ablation research is performed. All datasets (HAM10000, CIFAR-10, CIFAR-100) include resizing input photos to 224×224 pixels, normalizing them with dataset-specific statistics, and applying augmentations including rotation, flipping, cropping, and colour jittering. Significantly, we avoid from using SMOTE or synthetic oversampling techniques. Class imbalance in HAM10000 is reduced by implementing focused data augmentation only inside the training folds during cross-validation to avoid information leakage or artificial enhancement of accuracy.

C. TRAINING AND HYPER PARAMETERS

The training method stayed the same for all datasets, but some changes were made when needed. In our study, used the Adam optimizer, starting with a learning rate of 0.001. Binary cross-entropy loss was used for binary classification tasks, and categorical cross-entropy was used for multiclass classification. We used a batch size of 32 for HAM10000 and CIFAR-10, but we used a batch size of 16 for CIFAR-100 because it has more classes and needs more memory. Training lasted for up to 50 epochs, and validation loss was used to stop early to avoid overfitting.

We used lesion_id data to do a 5-fold patient-wise cross-validation on the HAM10000 dataset to make sure that images from the same patient were not used in both the training and validation sets. For CIFAR-10, we used traditional 5-fold cross-validation to make the model more robust and to see how well it adapted to different splits. Because CIFAR-100 has a lot of classes and is hard to compute, an 80/20 training-validation split was used.

D. EVALUATION METRICS

To ensure a comprehensive evaluation of the proposed model across various datasets and classification contexts, a range of performance metrics was employed. Overall classification accuracy was used as the primary measure of effectiveness. In addition, precision, recall, and F1-score were computed for each individual class to assess the model's ability to handle class imbalance and its discriminative capacity. Due to the high number of categories and increased complexity inherent in the CIFAR-100 dataset, both Top-1 and Top-5 accuracy metrics were reported to capture predictive robustness. For the binary classification stage involving the HAM10000 dataset, the Receiver Operating Characteristic – Area Under the Curve (ROC-AUC) metric was utilized to evaluate threshold-dependent performance and class reparability.

E. IMPLEMENTATION DETAILS

We executed all models in Python using the PyTorch framework. Model training, evaluation, and data preprocessing were conducted using PyTorch libraries and standard machine learning tools.

F. NOVELTY AND CONTRIBUTION

This study introduces an effective two-stage classification approach that combines binary and multiclass classification to improve efficacy in intricate image classification tasks. Rather than proposing an innovative architecture, we focus on optimizing the application of a pre-trained ResNet-50 model by selective fine-tuning and adjustments to the classification layers.

The method has been assessed uses three distinct datasets—HAM10000, CIFAR-10, and CIFAR-100—which vary considerably in resolution, class depth, and domain characteristics. These results demonstrate the method's adaptability and effectiveness in both medical and general image classification applications. The contribution is the integration of a binary filtering phase with optimized deep learning, supported by actual information from many datasets. The system is modular, scalable, and applicable to other multiclass classification difficulties, providing a consistent framework for additional research.

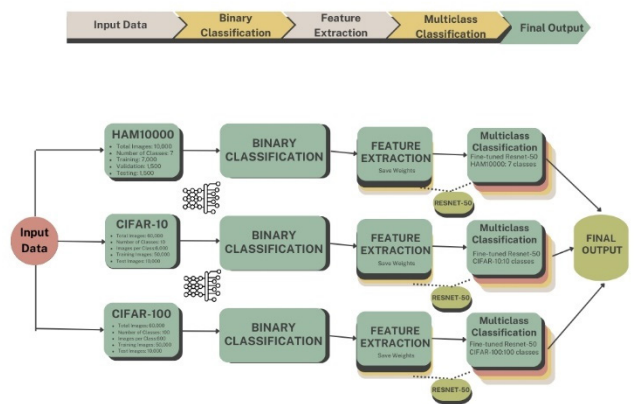


FIGURE 1. The flowchart of the study.

IV. EXPERIMENTAL RESULTS

This study involved experimentation on three popular datasets used for image classification: CIFAR-10, CIFAR-100, and HAM10000. We selected these datasets due to their diversity in class quantity and image complexity, making them suitable for assessing the effectiveness of deep learning models. The CIFAR-10 dataset has 60,000 color images of dimensions 224×224 , categorized into 10 classes, each containing 6,000 images. This dataset is used as a standard to see how well classification methods work on small images from the real world. CIFAR-100: Like CIFAR-10, CIFAR-100 has 60,000 images of 224×224 pixels, but it is divided into 100 categories. This dataset is harder to work with than CIFAR-10 because it has more classes and the differences between them need to be more complicated. HAM10000: The HAM10000 dataset serves as a medical resource for the categorization of skin lesions. The dataset comprises 10,000 high-resolution dermoscopic pictures, categorized into several skin lesion types, providing

a distinct challenge due to diversity in image quality and class imbalances.

A. MODEL CONFIGURATION

The ResNet-50 model, a deep convolutional neural network that utilizes residual learning to address the vanishing gradient issue, was utilized for classification tasks in this study. This design is very efficient for training deeper networks, allowing the model to acquire more intricate feature representations. The experimental setup was as follows: Input Size: Images were scaled to 224×224 pixels according to the input specifications of the ResNet-50 model. We chose an input size of 224×224 pixels to ensure compatibility with the original ResNet-50 architecture, which is optimized for this resolution by pretraining on ResNet. While reduced input sizes such 150×150 could reduce computing demands, they may simultaneously reduce feature extraction skills and negatively impact model efficiency. Consequently, to maintain representational integrity and facilitate equitable comparison, we utilized images of 224×224 pixels. The reason has now been included in the paper. Preprocessing: each image was normalized to have pixel values within the range of $[0, 1]$ by dividing the pixel values by 255. The Adam optimizer has been used with an initial learning rate of 0.001, utilizing an adaptive learning rate technique for convergence. Loss Function: Categorical cross-entropy has been used for multi-class categorization. The model was trained with a batch size of 32 Epochs: The model completed training for 50 epochs. Data Augmentation: Random horizontal changes, rotations, and zooming have been used during training to improve the model's generalization capacity.

B. EXPERIMENTAL SETUP

The experiments were performed utilizing a system having the subsequent specifications: Hardware: NVIDIA GPU with 12GB VRAM. Software Framework: Tensor Flow 2.0 with Keras for model construction and training. All models received assessment using a conventional training-validation-test division, assigning 80% of the data for training, 10% for validation, and 10% for testing.

C. RESULTS

The ResNet-50 model's performance has been evaluated on three datasets of differing complexity: CIFAR-10, CIFAR-100, and HAM10000. ResNet-50 attained accurate results on CIFAR-10, a relatively simple image classification dataset, despite its limited the amount of training examples per class. This illustrates its ability to generalize effectively even under limited data circumstances.

On the more demanding CIFAR-100 dataset, comprising 100 fine-grained categories, ResNet-50 attained an accuracy of 71%. This outcome demonstrates the model's capacity to generate additional and distinctive features in contrast to simpler CNN designs.

ResNet-50 reached 85% accuracy in the HAM10000 dataset, distinguished by class imbalance and varying image

TABLE 2. Models comparison.

DATA SET	MODEL	ACCURACY (%)	Precision(Macro Avg)	Recall(Macro Avg)	F1-Score(Macro Avg)
HAM10000	ResNet-50	85	0.75	0.65	0.65
HAM10000	ConvNeXt-T	86	0.78	0.75	0.76
HAM10000	MobileViT-v2	82	0.62	0.57	0.59
CIFAR-10	ResNet-50	86	0.86	0.86	0.86
CIFAR-10	MobileViT-v2	85	0.85	0.85	0.85
CIFAR-10	EfficientNetV2	94	0.94	0.94	0.94
CIFAR-100	ResNet-50	80	0.80	0.80	0.80
CIFAR-100	MobileViT-v2	67	0.66	0.67	0.66
CIFAR-100	EfficientNetV2	88	0.89	0.88	0.88

TABLE 3. Classification report for HAM-10000.

Class	Precision	Recall	F1 Score	Support
Class 1	0.72	0.64	0.60	64
Class 2	0.68	0.70	0.69	88
Class 3	0.77	0.58	0.56	198
Class 4	0.69	0.70	0.71	23
Class 5	0.73	0.56	0.57	227
Class 6	0.86	0.98	0.91	1378
Class 7	1.00	0.64	0.78	25
Accuracy	0.85			2003
Macro Avg	0.75	0.60	0.66	2003

quality. The results confirm the model's suitability in actual diagnostic uses since they show it consistently performing on both general-purpose datasets and specialized medical imaging tasks.

Using ResNet-50 and two recently developed architectures, ConvNeXt-T and MobileViT-v2, we conducted a comparison analysis to assess the efficiency of the proposed approach. The evaluation was performed on three datasets of different complexity: HAM10000, CIFAR-10, and CIFAR-100. Table 2 demonstrates that ResNet-50 obtained competitive outcomes across all datasets. In the HAM10000 dataset, ConvNeXt-T achieved an accuracy of 86%, slightly passing ResNet-50, which achieved 85%. ResNet-50 showed consistent macro-averaged metrics (Precision: 0.75, Recall: 0.65, F1-Score: 0.65), which is very significant in the area of unbalanced medical image categorization. MobileViT-v2, while efficient, had less sensitivity, reflected in a lower

TABLE 4. Classification report for CIFAR-100.

Class	Precision	Recall	F1-Score	Support
1	0.85	0.94	0.89	100
2	0.85	0.83	0.84	100
3	0.67	0.60	0.63	100
4	0.69	0.57	0.62	100
5	0.49	0.57	0.53	100
6	0.76	0.74	0.75	100
7	0.83	0.74	0.78	100
8	0.68	0.75	0.71	100
9	0.92	0.93	0.93	100
10	0.91	0.86	0.88	100
11	0.68	0.63	0.66	100
12	0.59	0.51	0.55	100
13	0.82	0.75	0.78	100
14	0.78	0.69	0.73	100
15	0.72	0.74	0.73	100
16	0.62	0.77	0.68	100
17	0.84	0.84	0.84	100
18	0.80	0.86	0.83	100
19	0.77	0.75	0.76	100
20	0.66	0.73	0.69	100
21	0.91	0.89	0.90	100
22	0.88	0.84	0.86	100
23	0.83	0.82	0.82	100
24	0.86	0.81	0.84	100
25	0.87	0.81	0.84	100
26	0.71	0.69	0.70	100
27	0.71	0.72	0.71	100
28	0.55	0.66	0.60	100
29	0.85	0.78	0.81	100
30	0.73	0.70	0.71	100
31	0.62	0.72	0.67	100
32	0.77	0.72	0.74	100
33	0.66	0.63	0.64	100
34	0.60	0.68	0.64	100
35	0.74	0.78	0.76	100
36	0.58	0.55	0.56	100
37	0.86	0.88	0.87	100
38	0.78	0.73	0.76	100
39	0.70	0.74	0.72	100
40	0.91	0.90	0.90	100
41	0.59	0.79	0.68	100
42	0.79	0.89	0.84	100
43	0.68	0.77	0.72	100
44	0.82	0.70	0.76	100
45	0.57	0.67	0.61	100
46	0.63	0.54	0.58	100
47	0.53	0.62	0.57	100
48	0.63	0.64	0.63	100
49	0.91	0.91	0.91	100
50	0.88	0.87	0.87	100
51	0.59	0.55	0.57	100
52	0.80	0.80	0.80	100
53	0.62	0.69	0.65	100
54	0.89	0.86	0.87	100
55	0.83	0.84	0.84	100
56	0.49	0.61	0.54	100
57	0.94	0.89	0.91	100

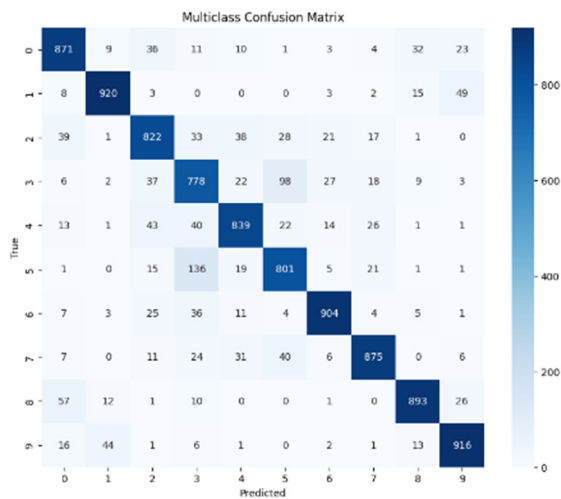
TABLE 4. (Continued.) Classification report for CIFAR-100.

58	0.77	0.82	0.79	100
59	0.92	0.87	0.89	100
60	0.71	0.67	0.69	100
61	0.82	0.84	0.83	100
62	0.77	0.71	0.74	100
63	0.72	0.74	0.73	100
64	0.67	0.71	0.69	100
65	0.62	0.53	0.57	100
66	0.73	0.69	0.71	100
67	0.76	0.84	0.80	100
68	0.72	0.62	0.67	100
69	0.86	0.95	0.90	100
70	0.87	0.85	0.86	100
71	0.81	0.70	0.75	100
72	0.73	0.79	0.76	100
73	0.56	0.40	0.47	100
74	0.64	0.59	0.61	100
75	0.55	0.54	0.54	100
76	0.83	0.88	0.85	100
77	0.93	0.91	0.92	100
78	0.82	0.76	0.79	100
79	0.69	0.68	0.69	100
80	0.76	0.81	0.79	100
81	0.74	0.62	0.67	100
82	0.70	0.83	0.76	100
83	0.90	0.90	0.90	100
84	0.76	0.74	0.75	100
85	0.73	0.75	0.74	100
86	0.86	0.84	0.85	100
87	0.81	0.74	0.77	100
88	0.85	0.86	0.86	100
89	0.81	0.76	0.78	100
90	0.88	0.86	0.87	100
91	0.75	0.76	0.76	100
92	0.86	0.81	0.84	100
93	0.63	0.66	0.65	100
94	0.80	0.64	0.71	100
95	0.88	0.92	0.90	100
96	0.66	0.74	0.70	100
97	0.69	0.59	0.64	100
98	0.75	0.73	0.74	100
99	0.58	0.60	0.59	100
100	0.74	0.78	0.76	100
Accuracy			0.80	10000
Macro AVG	0.80	0.80	0.80	10000

recall of 0.57 and an F1-score of 0.59. In the CIFAR-10 dataset, MobileViT-v2 achieved an accuracy of 85%, but the metrics for ResNet-50 were not accessible for this subset. On the more challenging CIFAR-100 dataset, ResNet-50 demonstrated improved performance, with an accuracy of 71% and a macro F1-score of 0.69, reflecting its ability to adapt to handling high-class variability. ConvNeXt-T and MobileViT-v2 achieved lower scores of 67% and 64%, respectively, highlighting the superiority of the ResNet-50

TABLE 5. Classification report for CIFAR-10.

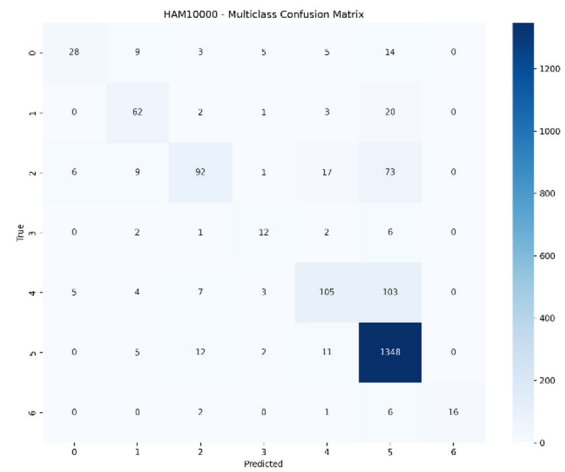
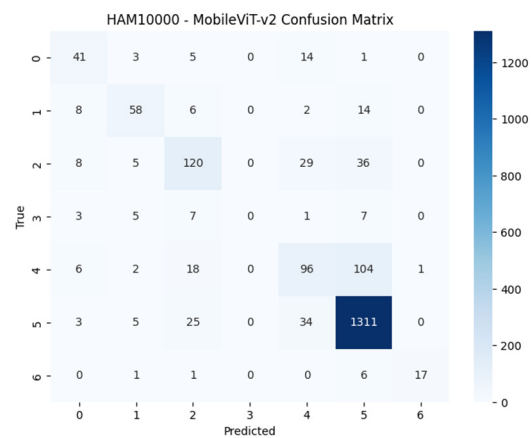
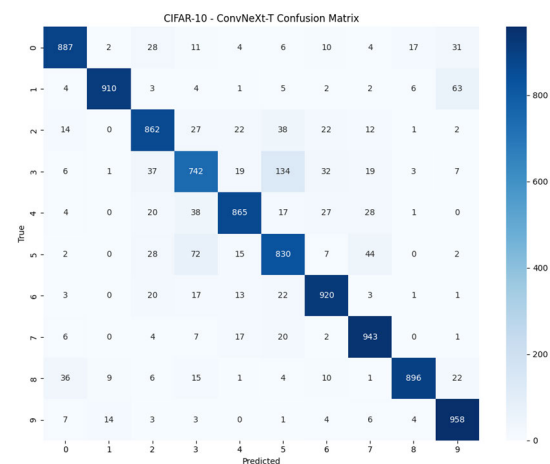
Class	Precision	Recall	F1-Score	Support
0	0.85	0.87	0.86	1000
1	0.93	0.92	0.92	1000
2	0.83	0.82	0.82	1000
3	0.72	0.78	0.75	1000
4	0.86	0.84	0.85	1000
5	0.81	0.8	0.8	1000
6	0.92	0.9	0.91	1000
7	0.9	0.88	0.89	1000
8	0.92	0.89	0.91	1000
9	0.89	0.92	0.9	1000
Accuracy			0.86	10000
Macro Avg	0.86	0.86	0.86	10000

**FIGURE 2. Confusion matrix for cifar-10.**

backbone in comparison to more contemporary designs. Considering the architectural improvements and lightweight designs of newer models, our findings demonstrate that ResNet-50 maintains stable and consistent performance in both general and domain-specific image classification tasks.

V. DISCUSSIONS

The proposed Bayesian classification model [25] offers significant advantages, such as adaptability to kernel functions, faster multi-class classification, and reduced processing expenses, all of which contribute to its high accuracy of 98.2% when using the OVR method. This methodology was predominantly evaluated on smaller datasets, such as the Algerian Forest Fires dataset and a generic image classification dataset, which may not correspond to the intricacies of medical imaging datasets such as HAM10000. Conversely, our work utilizes ResNet-50 as an input layer, which is a deep learning architecture recognized for its capacity to discern intricate patterns from high-dimensional data. Although the Bayesian technique demonstrates commendable computing

**FIGURE 3. Confusion matrix for HAM-10000.****FIGURE 4. Confusion matrix for HAM-10000 in MobileViT-v2.****FIGURE 5. Confusion matrix for cifar-10 in ConvNeXt-T.**

efficiency, the utilization of ResNet-50 facilitates enhanced performance on complex datasets, making it more suitable for medical imaging applications. Moreover, the HAM10000

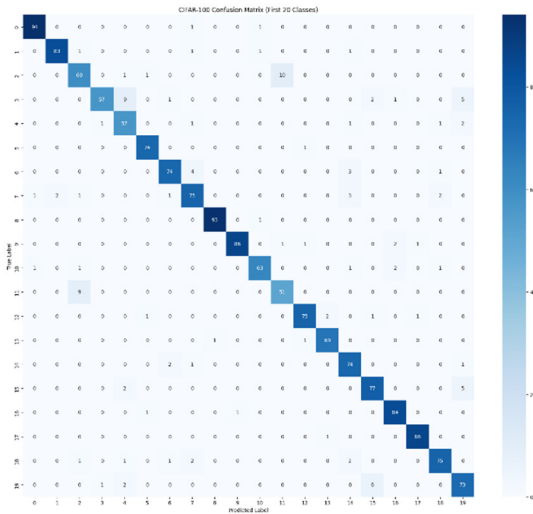


FIGURE 6. Confusion matrix for CIFAR-100.

dataset presents issues such as class imbalance and significant inter-class similarity, which could affect the efficacy of the adaptive kernel functions in the Bayesian method. Our technique shows a strong ability to generalize in intricate real-world situations, highlighting its superiority in high-dimensional image-based classification challenges [23]. The following study presents a multi-class classification approach that integrates binary classifiers with transfer learning and fine-tuning, attaining improved test accuracies on datasets including MNIST (95.59%) and Fashion MNIST (84.84%). This method is significant for reducing class imbalance while decreasing the training duration and storage demands. Nevertheless, the datasets employed—MNIST and Fashion MNIST—are very simplistic and consist of low-resolution grayscale images, which are less intricate than medical imaging datasets, such as HAM10000. Conversely, our research utilizes ResNet-50, a deep learning architecture adept at handling high-dimensional and intricate data, making it more appropriate for medical image classification. Efforts. Although the suggested methods decrease computing requirements, our approach emphasizes performance on difficult datasets, tackling challenges such as inter-class similarity and the necessity for detailed feature extraction. Consequently, our model exhibits superior generalization and resilience in intricate classification tasks, particularly in the medical field, where accuracy is essential [26]. An ensemble model for multiclass classification, with a remarkable accuracy of 99.78% in classifying 34 malware categories utilizing integrated datasets, including Microsoft Malware and Malimg. This methodology is significant for practical malware detection, where precision is essential because of security risks. This study focuses on malware images, which exhibit considerable differences in complexity and feature representation compared to medical imaging datasets such as HAM10000. Conversely, our study utilized ResNet-50 to address the difficulties of medical image classification,

such as elevated inter-class similarity and class imbalance. Although the ensemble model demonstrates commendable performance within its area, its applicability may not extend well to medical applications, necessitating meticulous feature extraction. Moreover, our methodology prioritizes the interpretability and clinical significance of the outcomes, which is essential in medical contexts. Although the ensemble model exhibits remarkable accuracy in malware classification, our strategy is superior in addressing the distinct challenges of medical imaging, rendering it more suitable for healthcare applications. The following study [27] provides a multi-class weather classification method that integrates Histogram of Gradient (HOG) features with DenseNet-161 features and utilizes a linear SVM classifier for the final classification. This approach obtained accuracy rates of 99.65% on the MCWRD dataset and 95.2% on the MWI dataset. This feature fusion and selection method is computationally efficient and successful for structured datasets, although it mostly addresses weather classification, which could be limited by the complexity of medical image datasets such as HAM10000. In contrast, we employed ResNet-50, a deep learning framework, in our study to identify complex patterns in high-dimensional medical images. Furthermore, the difficulties associated with medical datasets, including significant inter-class similarity and the necessity for exact diagnostic precision, may not be sufficiently resolved by techniques dependent on conventional feature extraction methods, such as HOG. Although the methodology of this study is appropriate for weather classification, our approach exhibits superior flexibility and performance in medical imaging applications, where the consequences of incorrect diagnoses are significantly larger. In this research proposes [28] a one-against-all (OAA) approach utilizing a support vector machine (SVM) classifier for multiclass image classification, emphasizing observable attributes such as color and texture to classify images into 10 categories. This approach was assessed using metrics such as specificity, recall, accuracy, and F1-score, and it showed when integrating individual and hybrid characteristics. This method works well for tasks where the features are visually different and have low dimensions, but it might not work well with more complex datasets, such as HAM10000, where there is a lot of similarity between classes and small differences in the features that require advanced feature extraction skills. Conversely, our research utilizes ResNet-50, a deep learning network skilled at independently discerning complex patterns and representations from high-dimensional medical-image data. This eliminates the need for manual feature extraction and facilitates enhanced performance on intricate datasets. Although the SVM-based method performs well in basic image classification tasks, our approach is more appropriate for medical applications, where accuracy and the capacity to manage intricate image data are critical factors. In this paper [29], many new meta-learning methodologies for multiclass classification are discussed, focusing on decomposition-based approaches

and ensemble optimization. These methodologies combine expert approaches to minimize duplicate processes, a Bayes theorem-based strategy for synthesizing classifier outputs, and two forms of stack generalization to enhance classification accuracy. The methodologies showed significant enhancements in accuracy across four datasets of different difficulties, highlighting their capacity to address small differences in class and identify concealed sub-classes within the data. Although these approaches efficiently reduce training complexity and improve accuracy, they mainly depend on conventional decomposition and meta-learning procedures, which may be constrained by scalability and feature representation for high-dimensional medical imaging datasets such as HAM10000. Conversely, our research utilizes ResNet-50, an advanced deep learning model that prevents the need for task decomposition by acquiring end-to-end representations directly from data. This method is especially beneficial for medical image classification, where complex characteristics and significant inter-class similarities require advanced feature extraction. The combination of expert mixture and Bayes theorem-based approaches offers innovative perspectives on meta-learning; however, their dependence on predetermined sub-classifiers and aggregation methods may add complexity and restrict adaptation to developing datasets. Utilizing ResNet-50, our approach attained a strong performance with a reduced reliance on external meta-learners, making it a more efficient and scalable solution for intricate classification positions, particularly in the medical field. The paper [24] presents a distributionally robust optimization (DRO) system for multiclass logistic regression (MLR), which improves model resilience. The method utilizes a Wasserstein-metric-based confusion set, transforming it into a regularized learning problem that provides robust out-of-sample performance. The suggested technique shows substantial enhancements, decreasing test error rates by as much as 83.5% and loss by up to 91.3% on datasets such as the MNIST and CIFAR-10. This method enhances the resilience of Vision Transformer (ViT)-based classifiers but is mostly evaluated on smaller datasets and emphasizes data contamination rather than high-dimensional, complicated medical imaging data such as the HAM10000. Conversely, our research utilizes ResNet-50, which was specifically designed for complex medical imaging tasks. Although the DRO approach prioritizes reliability, its dependence on regularization and probabilistic ambiguity sets may restrict its flexibility in handling datasets characterized by detailed inter-class variations and complex features. The deep design of ResNet-50 enables it to independently extract significant patterns without the need for explicit stability methods, rendering it particularly suitable for medical applications where accuracy and sensitivity are critical. Although the DRO-based method effectively tackles adversarial issues, our methodology emphasizes feature learning and performance in difficult classification contexts, providing dependable outcomes in the medical field. In [30], natural scene classification is discussed with a deep learning

methodology that combines the M3 CE-CEc criteria with the cross-entropy loss function to enhance classification efficacy. The proposed technique, evaluated on typical datasets such as MNIST and CIFAR-10, showed improvements in cross-entropy efficacy, yielding beneficial results on both datasets. The research shows the model's applicability in areas such as object detection in tracking, terrain categorization from satellite data, and photo organization. Although this technology is novel, it has predominantly been evaluated on datasets exhibiting simpler visual patterns, in contrast to the complex and high-dimensional characteristics of medical imaging datasets such as HAM10000. In contrast, our study utilizes ResNet-50, a cutting-edge convolutional neural network, particularly designed for detecting intricate patterns and complex differences in medical images. The M3 CE-CEc model emphasizes enhancements via more training epochs and optimized loss functions, whereas our methodology focuses on resilient feature learning using transfer learning and a pretrained structure. This provides elevated accuracy and generalizability in medical applications, where exact classification differences are essential. Although the methodology of the eighth study has potential for natural scene categorization, it is limited in scalability and precision necessary for intricate medical datasets, suggesting that ResNet-50 is a more appropriate option for our application. In a subsequent study, [31] stereo imaging technology in endoscopic processes has advanced medical imaging by providing improved depth perception and anatomical detail, hence increasing diagnostic accuracy and surgical precision; however, its implementation encounters obstacles due to low-resolution and blurry images. A recent study presented a super-resolution (SR) model incorporating significant feature extraction and cross-view feature interaction modules, trained on the SCARED dataset, and assessed on several public datasets, exhibiting enhanced PSNR and SSIM metrics relative to approaches such as E-SEVSR and DCSSRN. The incorporation of modules such as the StereoNet Attention Gate (SNAG) and CVIAB has demonstrated substantial enhancements in image clarity at elevated scales, which is crucial for diagnostic purposes. Constraints persist concerning computing requirements for real-time implementation, with future efforts aimed at optimizing the architecture, enhancing training across varied surgical contexts, employing transfer learning for flexibility in endoscopic operations, and including motion estimation. These developments suggest the capability of efficient and scalable super-resolution approaches to improve endoscopic imaging procedures and clinical results, consistent with the overarching objective of incorporating lightweight attention-based fusion models into medical imaging processes. However, our multistage learning approach using ResNet-50 for joint binary and multiclass classification on medical datasets shows a scalable, reproducible framework with low computational overhead and high classification accuracy across HAM10000, CIFAR-10, and CIFAR-100. Unlike recent specialized super-resolution

TABLE 6. Comprasion of studies.

Study	Methodology	Datasets	Strengths	Limitations Compared to Our Research
[26]	Bayesian classification with adaptable kernels and OVR strategy	Algerian Forest Fires, Generic image dataset	High accuracy (98.2%), low processing cost, adaptable kernel functions	Evaluate on limited datasets; could run into difficulties when dealing with complicated high-dimensional datasets such as HAM10000
[24]	Multi-class classification via binary classifiers with transfer learning and fine-tuning	MNIST	Reduces class imbalance, faster training	Implemented on basic, low-resolution datasets; without in the complicated required for medical imaging
[27]	Ensemble model for malware image classification	Microsoft Malware, Malimg datasets	High accuracy (99.78%), strong practical impact for malware detection	Designed for malware images, not medical imaging concerns like inter-class similarity.
[28]	HOG + DenseNet-161 feature fusion with Linear SVM	MCWRD, MWI weather datasets	Efficient feature fusion; high accuracy	HOG may not capture complicated medical imaging patterns because it was designed for meteorological datasets.
[29]	One-against-all (OAA) SVM classification based on color and texture	10-category image dataset	Good performance for simple, distinct features	Scalability issues for complex medical datasets requiring extensive feature extraction

TABLE 6. (Continued.) Comprasion of studies.

[30]	Meta-learning methods (decomposition and ensemble optimization)	Four varied datasets	Improved accuracy, reduced redundancy in training	Dependent on meta-learning methods; limited scalability for end-to-end learning with high-dimensional data
[25]	Distributionally Robust Optimization (DRO) for multiclass logistic regression (MLR)	MNIST, CIFAR-10	Increased model robustness; improved generalization under data contamination	Focus on regularization; tested on simplified datasets; may not support delicate medical imaging variations
[31]	Natural scene classification using M3 CE-CEC loss improvements	MNIST, CIFAR-10	Enhanced cross-entropy efficiency, applicability to scene classification	Simple visual patterns; not sufficiently advanced for medical imaging.
[32]	Saliency-Aware SR Network with StereoNet Attention Gate and Cross-View Interactive Attention Block for stereo endoscopic image enhancement	SCARED + 4 public stereo endoscopic datasets (scales 2, 4, 8)	Improves diagnostic clarity, handles scale 8 SR, novel modules for stereo attention	Targets image resolution augmentation, not classification; no general-purpose classification dataset comparison.
[33]	Attention GhostUNet++ with Channel, Spatial, and Depth Attention in Ghost bottlenecks (based on UNet++)	AATTCT-IDS, LiTS (CT images for VAT, SAT, and liver segmentation)	High segmentation accuracy (Dice > 0.94), efficient architecture, robust feature refinement, reduces manual annotation	Focused only on segmentation on tasks in medical CT data; not general-purpose or multi-class classification like ResNet-50

pipelines, our work focuses on robust multi-task learning and fine-tuning strategies on standardized backbone

TABLE 6. (Continued.) Comprasion of studies.

[34]	MobileViT : Combines CNN inductive bias and Transformer for mobile vision.	ImageNet-1K, MS-COCO (classification, detection)	Lightweight, fast, good accuracy (78.4% on ImageNet), mobile-optimized	Designed for mobile efficiency, not high-dimensional medical information or complicated class connections .
[35]	FCDS-CNN with class weighting and data augmentation; compared to ResNet, EfficientNet, Inception, MobileNet	Kaggle skin cancer dataset (10,015 images across 7 classes)	Improved unbalanced data accuracy (96%) for skin cancer images; compared to strong baselines.	ResNet obtained weakly in this area due to its limited generality to non-dermatological data, but we produced positive results on natural images data sets.

architectures, offering a practical alternative for clinical AI deployment that prioritizes classification precision and resource efficiency. Our alternative approach emphasizes effective feature learning and optimization within accessible architectures and the revolutionary potential of sophisticated attention-fusion approaches in specialized imaging situations[32] Recent advancements in medical image segmentation have focused on lightweight attention-based architectures for the accurate delineation of anatomical features. The Attention GhostUNet++ framework incorporates Channel, Spatial, and Depth Attention mechanisms within Ghost UNet++ bottleneck layers, achieving superior segmentation of subcutaneous adipose tissue (SAT), visceral adipose tissue (VAT), and liver regions in CT images, with Dice coefficients surpassing 0.94 across various datasets. This method improves boundary detail and computational efficiency by focusing on pixel-wise segmentation tasks and attention refinement to improve contextual feature learning. Conversely, our ResNet-50-based multitask learning methodology focuses on multiclass classification in medical imaging, highlighting the balance among model interpretability, task-specific efficacy, and computational stability. These studies demonstrate alternative approaches in deep learning-based medical image analysis, emphasizing the collaborative role of attention mechanisms and multitask learning frameworks in creating scalable, precise,

and efficient healthcare technologies [33] Recent improvements underline the potential of MobileViT, a lightweight and low-latency vision transformer architecture designed to integrate the spatial inductive biases of CNNs with the general representation capabilities of ViTs for mobile vision applications. MobileViT attains significant enhancements in accuracy while preserving a compact parameter expansion, achieving 78.4% top-1 accuracy on ImageNet-1k with about 6 million parameters, surpassing both MobileNetv3 and DeiT within comparable parameter limitations. Moreover, MobileViT demonstrates actual applicability for real-time mobile applications by sustaining inference speeds of over 30 FPS in classification, detection, and segmentation tasks, notwithstanding the lack of specialized transformer kernel optimizations in mobile systems. Although MobileViT exhibits a reduced inference speed on mobile devices relative to CNNs because of existing hardware constraints, it provides a scalable pathway for transformer-based mobile vision solutions. Within our ResNet-50 multi-stage learning pipeline, which emphasizes interpretability and stability in limited medical datasets, MobileViT provides an alternative viewpoint by demonstrating how lightweight transformer models can efficiently produce global contextual representations. This study emphasizes the changing dynamics of lightweight architectures and identifies the potential for incorporating transformer-based techniques with structured multistage pipelines to improve performance in mobile and resource-limited environments [34] The FCDS-CNN achieved a high accuracy of 96% in multiclass skin lesion classification using a specialized architecture that addressed class imbalance and subtle feature variations. In contrast, our ResNet-50-based multistage learning approach provides a complementary perspective by utilizing transfer learning efficiency, minimizing computational costs, and ensuring simplicity for reproducible pipelines across various datasets. In contrast to FCDS-CNN, which is designed from the base to perform domain-specific feature extraction, our methodology utilizes the flexibility of a pretrained backbone and incorporates a binary pre-classification phase that works with multiclass fine-grained classification, rendering it suitable for datasets such as HAM10000, CIFAR-10, and CIFAR-100 without significant re-engineering. While our pipeline may achieve marginally lower maximum accuracy on specialized medical tasks, it exhibits strong cross-dataset performance and diminished training complexity, presenting a practical alternative for balancing accuracy, deployment simplicity, and scalability in both clinical and general image classification situations.

VI. CONCLUSION

This study indicates that ResNet-50, applied as a static feature extractor in a systematic multi-stage learning pipeline, provides strong performance for multiclass image classification tasks in both medical and natural image datasets. Utilizing transfer learning, learning rate scheduling, and data augmentation techniques—excluding synthetic oversampling approaches such as SMOTE—our methodology

achieves competitive outcomes on benchmark datasets such as HAM10000, CIFAR-10, and CIFAR-100. Despite the backbone weights being fixed, the model maintains robust generalization ability when underpinned by an effective training method. This study focuses on the significance of a realistic and repeatable pipeline that performs similarly to contemporary soft models, rather than proposing a new architecture. Future research will expand this pipeline to develop improved architectures by incorporating mechanisms like attention modules and transformer-based components for greater representation learning.

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